Characteristics from a graph

1

a Yes

b No

2

a They become very large in the negative direction.

b They become very large in the positive direction.

3

a

i It has an absolute minimum.

ii Yes, there is a local minimum.

b

i No, it does not have either.

ii Yes, it has local maximums and minimums.

c

i Yes, it has an absolute maximum.

ii Yes, there are local maximums and minimums.

4

a

i $(3,0)$ **ii** $(0,-3)$

b

i $(4,0)$ **ii** $(0,4)$

c

i $(-1,0)$ **ii** $(0,-2)$

5

a

i $(-8,0),(0,0)$

ii $(0,0)$

iii They become very large in the positive direction

iv They become very large in the positive direction

b

i $(-2,0),(-4,0),(3,0)$

ii $(0,5)$

iii They become very large in the negative direction

iv They become very large in the positive direction

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c

i The function does not have x-intercepts.

ii $(0, 1)$

iii They become very large in the positive direction

iv They approach zero

6

a The function does not have x-intercepts.

b $(0, 1)$

c They become very large in the positive direction

7 $(-2, 0)$

8

a 0 km **b** 30 km **c** 1 hour

9 1 km

10 $\frac{3}{2} < x < 2$

11

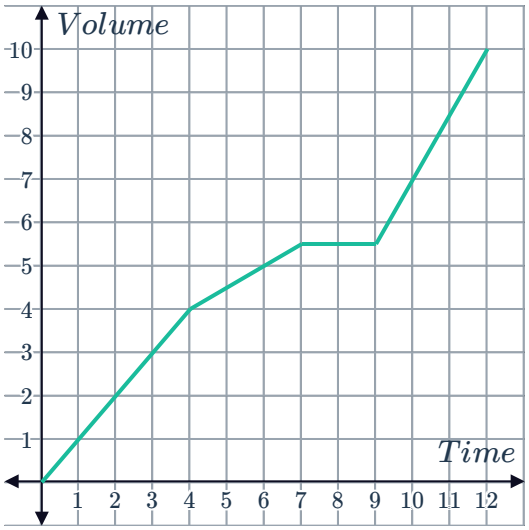
a No **b** No **c** Yes **d** No

12 $5 \leq t \leq 6$

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13

a



b $\frac{5}{6}$ million gallons of water per hour

Additional questions

14

a

i Domain: $-2 < x \leq 4$

ii Range: $-5 < y \leq 1$

iii x-intercept is 3
y-intercept is -3

iv Rate of change is 1

b

i Domain: all real values

ii Range: all real values

iii x-intercept is 3
y-intercept is 6

iv Rate of change is -2

c

i Domain: all real values

ii Range: $y = 2$

iii There is no x-intercept
y-intercept is 2

iv Rate of change is 0

d

i Domain: $-2 \leq x \leq 4$

ii Range: $-2 \leq y \leq 1$

iii x-intercept is 2
y-intercept is -1

iv Rate of change is $\frac{1}{2}$

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15

a $(0, 9)$

b The height from which the soccer ball was kicked.

c $(3, 0)$

d The time at which the soccer ball hits the ground.

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